

Assignment: DNS Server Configuration in Windows Server

1. Install the DNS Server Role

Steps:

1. Open **Server Manager**.
2. Click **Manage → Add Roles and Features**.
3. Click **Next** until you reach the **Server Roles** page.
4. Select **DNS Server**.
5. When prompted, click **Add Features**.
6. Click **Next** until you reach the **Confirmation** page.
7. Click **Install**.
8. Wait for the installation to complete.
9. Click **Close**.

Result:

The **DNS Server role** is installed successfully on Windows Server.

2. Create a Forward Lookup Zone and Add an A Record

Steps:

1. Open **Server Manager → Tools → DNS**.
2. Expand the **Server Name**.
3. Right-click **Forward Lookup Zones** and select **New Zone**.
4. Click **Next**.
5. Select **Primary Zone** and click **Next**.
6. Enter the Zone Name as:

foodies.local

7. Keep the default settings and click **Finish**.
8. Right-click the **foodies.local** zone.

9. Select **New Host (A or AAAA)**.

10. Enter the following details:

Name: zomato

IP Address: 192.168.1.50

11. Click **Add Host**.

12. Click **Done**.

Result:

The **foodies.local** Forward Lookup Zone is successfully created, and an **A record** is configured:

zomato.foodies.local → 192.168.1.50

3. Create a CNAME Record

Steps:

1. Right-click the **foodies.local** zone.
2. Select **New Alias (CNAME)**.
3. Enter the following details:

Alias Name: swiggy

Fully Qualified Domain Name (FQDN) for Target Host: zomato.foodies.local

4. Click **Browse** to select the target host if required, or enter it manually.
5. Click **OK**.

Result:

A CNAME record is successfully created:

swiggy.foodies.local → zomato.foodies.local

4. Verify DNS Records Using NSLOOKUP

Steps:

1. Open **Command Prompt** on the client computer.

2. Run the following commands:

```
nslookup zomato.foodies.local
```

```
nslookup swiggy.foodies.local
```

Expected Result:

For the A record:

```
nslookup zomato.foodies.local
```

The command should return the IP address:

192.168.1.50

For the CNAME record:

```
nslookup swiggy.foodies.local
```

The command should show that **swiggy.foodies.local** is an alias of **zomato.foodies.local** and resolves to:

192.168.1.50

5. Simulate DNS Resolution

Commands:

Run the following commands from the client computer:

```
nslookup zomato.foodies.local
```

```
nslookup swiggy.foodies.local
```

Observation:

1. **nslookup zomato.foodies.local**

- The hostname is an **A record**.
- It directly resolves to the IP address **192.168.1.50**.

2. **nslookup swiggy.foodies.local**

- The hostname is a **CNAME record**.
- It first resolves **swiggy.foodies.local** to **zomato.foodies.local**.

- The target hostname then resolves to the IP address **192.168.1.50**.

DNS Resolution Flow:

zomato.foodies.local → 192.168.1.50

swiggy.foodies.local → zomato.foodies.local → 192.168.1.50

